Muon Scattering Tomography in Imaging Hidden Dense Materials

October 13, 2026



Endri Çelçima, Erind Çelçima, Enkel Ramollari, Sarah Temali, Leyla Sena Aksel, Luis Shella, Luka Shella, Klevi Murrizi, Marion Merko, Agron Mihasi

Turgut Ozal High School, Tirana

Mentors: Kastriot Capa, Nuray Donmez

Team: Breaking Beams

1 Introduction

Muon tomography is a technique that probes the internal structure of objects without physically opening them. This method relies on measuring the change in trajectory of muons as they pass through different materials. Because muons are relatively massive (a mass of approximately $105 MeV/c^2$), they lose energy slowly and can penetrate large thicknesses of material while still exhibiting measurable scattering when encountering dense elements.

In our experiment, we will try to figure out the internal structure of 3 test structures, as shown in Figure 1 and Table 1.

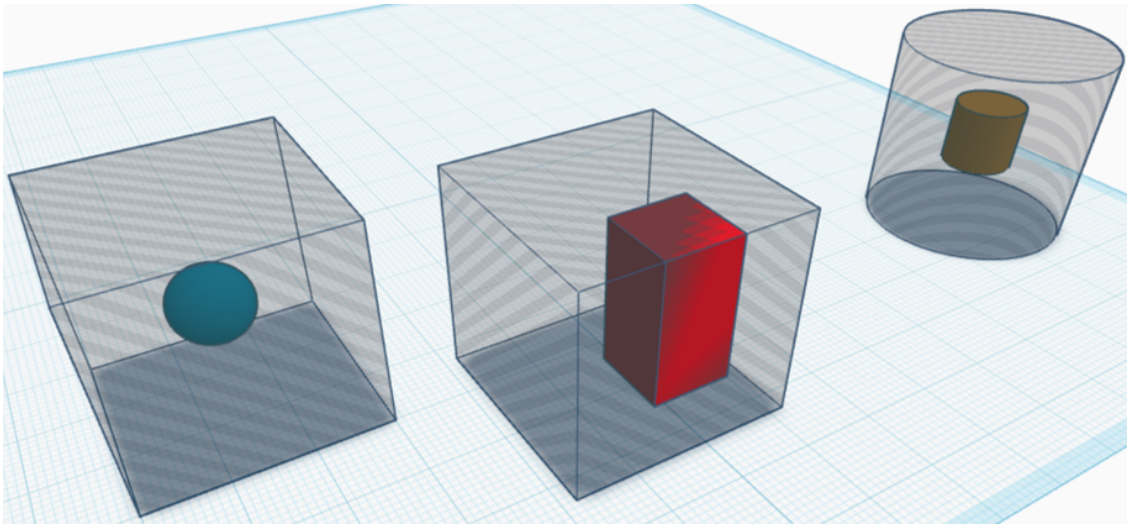


Figure 1 The test structures include: a hollow cube with a sphere placed in the center, one with a cuboid placed randomly and a last one with a cylindrical body and a cylinder inside, representing nuclear waste.$

Test Structure	Shape	Width	Length	Height	Internal Shape	Width	Length	Height	Metal
1	Cube	10	10	10	Sphere	6	-	-	Lead
2	Cube	10	10	10	Cuboid	4	4	8	Tungsten
3	Cylinder	10	-	12	Cylinder	4	-	6	Bismuth

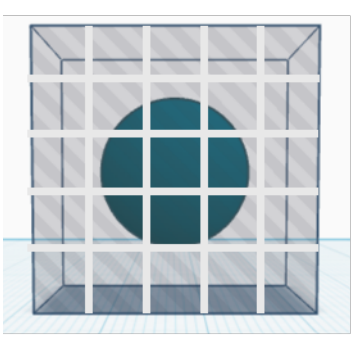
Table 1 Test structure details

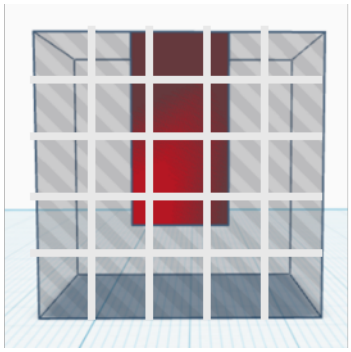
We will also propose a way to figure out the type of metal inside each test structure. Lead, Tungsten and Bismuth were chosen because they have high Z and distinguishable X_o. The shapes of the structure were chosen to show that different shapes can all be measured.

2 Methodology and Theoretical Framework

The first stage of the experiment is figuring out the shape of the internal structures. We will take advantage of the 2cm diameter of the T9 beam. We will divide the faces of each test structure into

2cm-by-2cm square segments and measure each independently, as shown on Figure 2.1, Figure 2.2 and Figure 2.3.





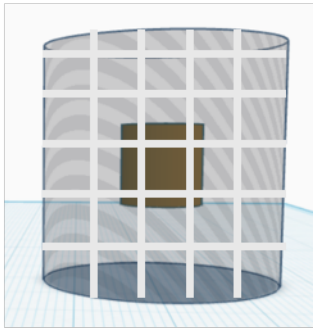


Figure 2.1, Figure 2.2, Figure 2.3 These are just visuals to better understand the separation, not to scale.

After each square segment of a face is completed, all the scatter angles for that face will be shown in a heatmap. After a face is completed, we will flip to another face and repeat. For structure 1 and 2, only 3 independent faces are needed, and for simplicity, we will do the same for 3.

The second stage of the experiment is figuring out the type of metal of the internal structures.

Characteristics of the scattering angle distribution are determined by material properties and muon momentum as shown in Highland:

$$\sigma_\theta = \frac{13.6MeV}{\beta cp}\sqrt{\frac{X}{X_0}}[1 + 0.038\ln(\frac{X}{X_0})]$$

Where X_0 is the radiation length of the material, X is the thickness of the material (which we will find experimentally) and $\beta = v/c$, which is ≈ 1 for these relativistic muons. X_0 can be expressed in g/cm^2 or cm. In this formula, we use it in cm. To convert from g/cm^2 to cm we divide by the metal's density:

$$X_0(cm) = \frac{X_0(/cm^2)}{\rho}$$

Since X_0 is specific for each metal, we can also figure out the type of metal inside, given that we know the other variables. Important information to note about each metal testes is given in Table 2.1.

Metal	Z	Density(g/cm^2)	Radiation length (X_0)(cm)
Lead(Pb)	82	11.34	0.56
Bismuth(Bi)	83	9.78	0.0
Lead(Pb)	74	19.25	0.35

Table 2.1

To measure the scattering angle, we first need the initial and final angles. We can find these by detector measurements; two coordinates are needed for each trajectory. Given the experimentally found coordinates (from the detectors) for the initial trajectory and for the final trajectory, we calculate the projected angles in x and y like so:

$$\theta_x = \arctan(\frac{dx}{dy})$$

$$\theta_y = \arctan(\frac{dy}{dz})$$

3 Experimental Setup

The experiment will take place at the PS Test Beam Facility at CERN. Our experimental setup is shown in Figure 3.1.

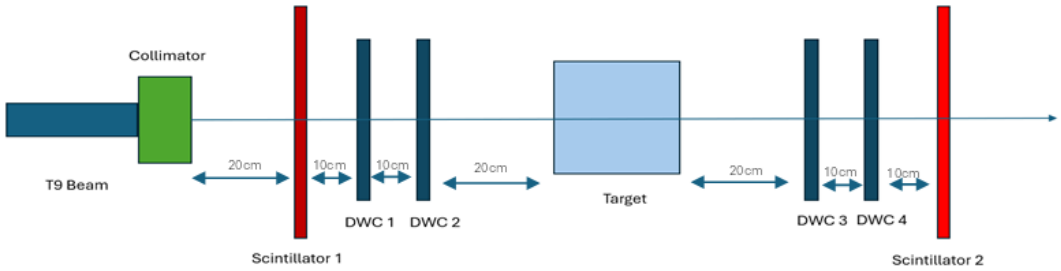


Figure 3.1 Experimental setup

In front of the T9 beam there will be a collimator, to filter other particles and allow only muons. Here we un into a limitation, since the muons will be scattered from the collimator and will not be concentrated. We look to CERN for a possible solution. Because of that, the 20cm distance from the collimator to Scintillator 1 may be altered accordingly. The beam will supply $3GeV/c$ momentum to the muons. Since we need a lot of measurements for accurate data, many spills will be used for each test structure. DWC 1 and DWC 2, as well as DWC 3 and DWC 4 will be placed 10cm apart, and this will be used for the z-coordinates. Each DWC will record its corresponding position coordinates measured in the theoretical framework. The detectors will be connected electronically in a scintillator trigger system to isolate individual muon events. Note that the setup will be altered for each segment of each test structure.

4 Data Analysis

After all trials are finished for each segment, the calculated angles will all be put in a histogram. We expect each histogram to have a bell-curve (Gaussian) shape centered at zero, like in Figure 4.1. The width of that bell curve is θ_0.

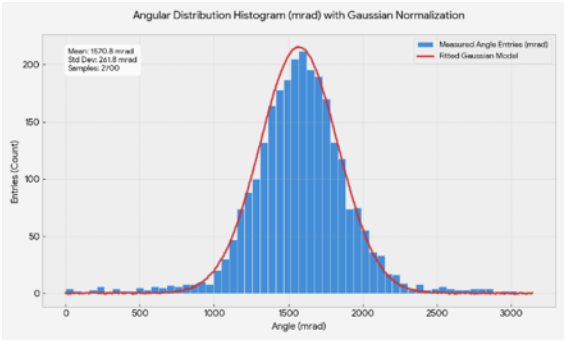


Figure 4.1 Image of expected histogram

This will be done for each face segment. After that is finished, we will plot a heatmap of θ_0 and the dimensions of each face, like in Figure 4.2.

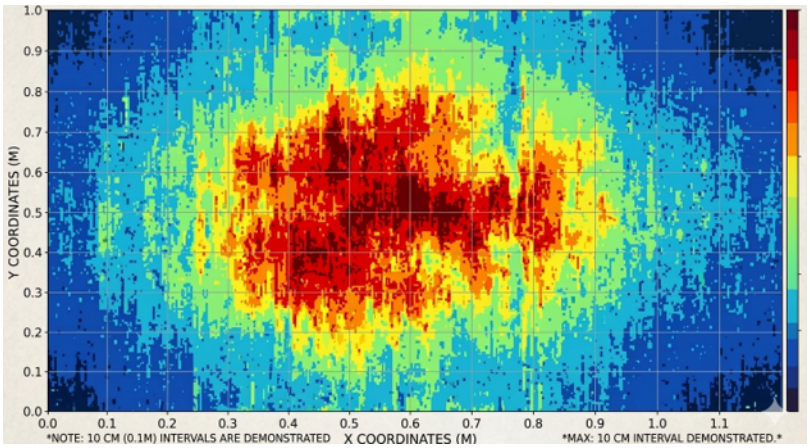


Figure 4.2 Image of expected heatmap of test structure 1

Using these heatmaps, the approximate 3D image can be constructed. We will arrange them in the positions of their corresponding faces. We would use the x, y and z coordinates of the heatmaps for the reconstructions. In regions where any faces show no scattering, we have no internal structures found. In regions where all faces show angles, we have an internal structure inside. The dimensions of these internal structures can be measured, since the heatmaps are constructed with the exact dimensions of the test structures in mind. This is shown in Figure 4.3 and Figure 4.4 From this, the Highland formula can be used to figure out the type of metal inside. This method can also be used if we have the opposite of these test structures, so a thick structure with a hollow part in the middle.

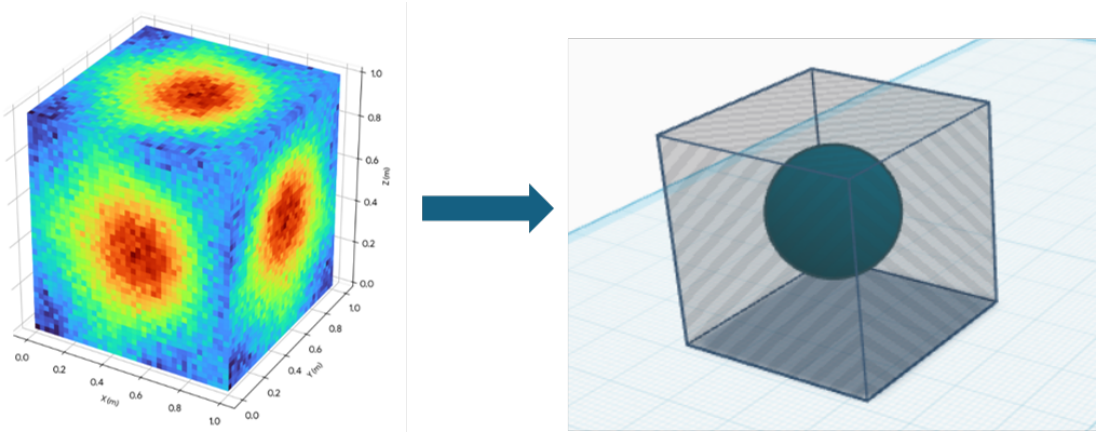


Figure 4.3 shows the expected heatmap reconstructon of test structure 1, and Figure 4.4 shows the 3D model of test structure 1.

We would like CERNs help for the propriate programs and construction of test structures. We realize that fully hollow structures are complicated, so an appropriate substitute like foam can be used.

5 References

1. https://journals.aps.org/prd/abstract/10.1103/PhysRevD.106.092003?utm_source=chatgpt.com
2. https://www.sciencedirect.com/science/article/pii/S0306454923005595#f0010
3. https://www.sciencedirect.com/science/article/abs/pii/0029554X75907430